

FROM THE
"DOFS - YOU BLEW IT STEVE"
 DEPARTMENT
 OR - "WATCH OUT FOR '...' IN WORDSTAR!"

Editor's Note:

The following article which appeared in our last issue was (obviously) missing a few (important) lines. Jim brought this to my attention at the last meeting so I went searching for the reason why ... I had reproduced the article from a text file which Jim had sent me, then merged it into the Wordstar file for the Newsletter. Apart from some minor alterations for the layout, I had not made any changes. A quick check of my Newsletter file showed all the missing lines - INTACT! They did, however have one thing in common, they all started with the initial characters ("_ _"). I didn't realise it at the time, but the "IGNORE" (.IG) command, which is used to put COMMENTS into the file can be abbreviated to (".."). In other words, WORDSTAR had done exactly as it was told, and IGNOREd some vital lines of my file. I have now re-edited the file, and reproduce it below:- (Steve :Ed)

FORMATTED PRINT OUT OF BASIC FILES

BY JIM OLIVER

The programme is presented here in the form it produces itself when 'working on' any file.

```

10 REM Two column print out of 110 LN=LN+1
.. Basic 120 LINE INPUT #1,L$(LN)
.. 130 E=EOF(1):IF E THEN 200
20 REM Files saved in ASCII fo 140 IF LEN (L$(LN)) <= ML THEN
.. rmat 150 L$(LN+1)=".. "+MID$(L$(LN)
.. 160 L$(LN)=LEFT$(L$(LN),ML)
30 REM D.Rogers./J.A.Oliver 22 170 LN=LN+1
.. 180 IF LN<=PL THEN 140
.. 190 IF LN<2*PL THEN 110
.. 200 P=INT(LN/2)
40 CLS:REM This is version No. 210 FOR I=1 TO P
.. 9 (PL=56) 220 LPRINT L$(I);TAB(C);L$(I+
.. AGE;ML=MAX CHARS/COL 230 NEXT I
70 LPRINTCHR$(15) 240 LPRINT CHR$(12)
80 PRINTTAB(4)"ENTER NAME OF F 270 IF LN=2*PL THEN LN=0 ELSE
.. ILE TO BE PRINTED":PRINT:PR 280 IF NOT E THEN 110
.. INT:INPUT F$ 290 CLOSE:END
90 OPEN"1:"+F$ FOR INPUT AS #
.. 1
100 LN=0

```

Sometimes it is convenient to print out your program listings in double columns. Most basic programmes have lots of waste space on the right hand side of the page so this both saves paper and makes it easier to follow the logic if the whole thing is on one page. The program given above is one originally developed by Don Rogers in response to my request for something in basic that would do the same thing as the CP/M PD file called DBL-EP.COM. Don came up with the ideas and I got them working!

The program which you want to print out must be on the same disc and be in ASCII format. (SAVE "1:file",A)

It was produced in this form by writing the strings (lines 220,240 etc) to a disc file instead of to the printer. This ASCII file was then transferred to CP/M. The variables were altered as follows, PL=22, ML=30, C=35. You can of course print to the screen, in case you wish to cram as much as possible onto it. Neat eh ?

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AND ANOTHER 'OOPS' !

SPECTRAVIDEO DISC DIRECTORY UTILITY.

BY JIM OLIVER

Editors Note:

(IN THE LAST ISSUE, I forgot to make one important correction which Jim had advised to me before going to press for this program. Namely, the inclusion of the sorting array (S(n)) in the initialisation routines in line 30. So I now reproduce that part of the listing, including the correction.

Sorry folks!! - Steve H)

```

10 REM DISC DIRECTORY READ,SAVE & PRINT
20 REM J.A.OLIVER JUNE87
30 SCREEN,0:COLOR1,10:CLS:MAXFILES=2:CLEAR9000:DIMF$(360),DN$(40)
  S(20):T=1:L=0:C=0:IV$=CHR$(27)+"p": NV$ = CHR$(27)+"q":
  CLOSE: ON STOP GOSUB 200:STOP ON
40 PRINT"
50 PRINT"          I DISC DIRECTORY I
60 PRINT"          ++++++
70 LOCATE10,7:PRINT"1: CREATE DISC DIRECTORY":PRINT:PRINTTAB(10);
  "2: PRINT OUT DIRECTORY":PRINT:PRINTTAB(10);"3: VIEW DIRECTORY
  ":PRINT:PRINTTAB(10)"4: QUIT":PRINT:PRINTTAB(10)"Please select
  a number":INPUT S:ON S GOTO 80,220,340,200
80 PRINT"Log in;";IV$+" which drive 1 or 2 "+NV$;" ";DS=INPUT$(1
  ):PRINTDS:D=VAL(D$):PRINT:INPUT "Disc name or no.":DN$:PRINT:I
  NPUT "(B)asic or (C)P/M Disc";CH$:ON INSTR("BbCc",CH$) GOTO
  90,90,400,400:PRINT"Please use 'B' or 'C' keys":GOTO80
  - - - and so on. See Last Issue
  //////////////////////////////////

```

.. Also did u know that u can rename files using WS so that CP/M .. will not recognise them? I've used all kinds of illegal .. letters like ! & * etc.

+++++

QUESTION + ANSWER

Why does this not print anything ?

```
INPUT AS(A):A=A+1:PRINT AS(A)
```

whereas :

```
INPUT AS(A):PRINT AS(A):A=A+1
```

DOES !

The above question from a member brings to mind the problems we all experience when 'DEBUGGING' any new program. This job, of finding out "What went wrong...?" is made harder by using a line with multiple statements, as above. If you are unsure of any such line, or as in this case KNOW that there is an error but "It's not obvious!", my recommendation is to break the line up .. often, this makes the error glaringly obvious. Let's try it here:-

Firstly, what are the variables, and pieces of Data that we're dealing with? -

- AS(n) is a string array, initialised with ten blank elements, and used to store input for later printing.
- A - is a numeric variable, used as a 'subscript', or as a pointer to the appropriate element within the array.

~~Secondly let's look at the statement which does work, in the order in which it would be processed by the computer.~~

1. The user inputs a string into the 'current' array element
2. The current array element is printed (ie what was just input)
3. The 'pointer' is incremented so that the next element is 'current'.

Finally, let's see where the other statement goes wrong:-

1. The user inputs a string into the 'current' array element
2. The 'pointer' is incremented to the 'NEW current element'
3. The NEW current element is printed (ie the element IMMEDIATELY AFTER the one most recently INPUT)

The above statement could therefore face two possible errors. Either the computer IS printing what it is told, ie a 'NULL' string (remember that a string array is 'initialised' with each element blank [or NULL]). OR, the string array subscript ('A') is pointing to a non-existent array element - which will probably give an 'OUT OF RANGE' error message, (ie unless an array is DIMENSIONED at the start of a program, the maximum number of elements that it may have is TEN).

This may seem a long-winded way of checking things out, but it doesn't take too long if you do it first as a 'mental exercise', following a program with a 'BUG' through step-by-step. Most bugs can at least be 'narrowed down' to a line, or a few lines of program that can be given 'The Treatment'.

ANOTHER QUESTION ...

DISC SPIN OFF PATCH

The well known disc spin off patch is fine for all normal disc use, but I have found that the fast disc turn off is the source of all my ASCII file errors. Often when creating files and addressing both drives the discs stop spinning between successive write cycles. All this turning on and off creates errors in the files. By booting the system on an unmodified disc (lucky I had one) the discs spin for a minute or so but all the errors disappear! SO SVI are not so silly after all.

Question : Does anyone know how to modify the disc spin off for, say, a 10 - 15 second spin off. The modified CP/M by Steve McNamee does this, but how? I've tried experimenting with the bytes and 'hang' the computer every time.

While I'm on the modified CP/M - did you know that Steve knows how to detect when there is a disc in the drive? You didn't? Well, try booting the system and then format a disc without a disc in the drive and see what happens. Clever chap that Steve.

***** SCREEN EDITOR

There are a few screen editors around, however I wrote the one ~~below to fill a need not catered for so far on SV.~~

Most new programs require instructions of some sort and incorporating these into a Basic listing can be time consuming. The following program will, on pressing the ESC key, write an ASCII file to disc (or tape by deleting the '1:') which contains all the text off the screen but written into Basic program lines which can be merged or appended into your listing. When RUN this ASCII file will faithfully display the original screen image. This avoids the lengthy trial and error methods I sometimes use.

If you want to move a line or word to the right, position the cursor at the left hand end of the line and press the '[' key. Use the ']' key to push it back again if required. I don't know why this process leaves a cursor image on the screen. To get rid of this simply position the moving cursor over the top of the unwanted cursor character (both will now disappear) and hit the space bar. The programme will accept both ASCII and graphics characters for those flashy titles etc.

Before saving the file with the ESC key move the cursor down to the function key line. If you don't an 'i' will be included in your file where ever the cursor was when ESC was pressed. This is the 'i' of Suzuki in the character set, why this happens is beyond me. The character is NOT CHR\$(105) as you will find out. To speed up the process of reading the video ram etc, the lines should be compacted after you understand whats going on.

```

0 REM screen editor J.A.Olive 360 VP=VPEEK(S+T):IF VP>95 T
.. r Nov.'87 .. HEN VP=VP-32 ELSE VP=VP+32
20 ' 370 AS(R)=AS(R)+CHR$(VP)
30 REM initialise 380 NEXT S
40 ' 390 IF AS(R)=STRING$(40," "
50 CLEAR 2000:DIM AS(23):L=0:W .. ) THEN AS(R)="PRINT" ELSE
.. =0:LN=0 .. AS(R)="PRINT"+CHR$(34)+AS(
60 DEF FN VR(X,Y)=(Y+40)+X .. R)+CHR$(34)+CHR$(59)
70 ON STOP GOSUB 580:STOP ON 400 R=R+1
80 SCREEN,0:WIDTH 40:CLS:PRINT 410 NEXT T
.. TAB(7)"SCREEN IMAGE TO DAT 420 '
.. A LINES":PRINT 430 ' save AS(R) to disc as AS
90 ' .. CII file
100 REM save function keys 440 '
110 ' 450 OPEN F$ FOR OUTPUT AS#1
120 ' FORT=&HFA1E TO &HFA1E+15 460 FOR T=0 TO R-1
.. 9 470 PRINT#1,STR$(LN);" ";AS
30 FK$=FK$+CHR$(PEEK(T)) .. (T)
40 POKE T,0 480 LN=LN+10
50 NEXT T 490 NEXT T
60 ' 500 CLOSE
70 INPUT "FILE NAME: ";F$:PRIN 510 RESTORE 530
.. T:PRINT:INPUT"WHICH DRIVE" 520 FOR T=932 TO 947:READ AS:V
.. ;DS:F$=DS+": "+F$ .. POKE T,ASC(AS)+64:NEXT T
80 PRINT:PRINT:INPUT"FIRST LI 530 DATA *,F,I,L,E,*,*,*,C,R,E
.. NE NUMBER";LN .. ,A,T,E,D,*
90 PRINT:PRINT:INPUT"DO 540 RES=INPUT$(1)
.. YOU WANT GUIDE LINES";GL$ 550 '
.. IF INSTR("yY",GL$)>0 THEN 560 REM restore function keys
.. VPOKE 2048,255-ELSE-VPOKE 570 '
.. 2048,0 580 SCREEN,1:FOR A=&HFA1ETO&HF
00 ' .. A1E+159
20 REM input to screen 590 POKE A,ASC(MID$(FK$,A-&HFA
.. 1E+1,1))
30 CLS 600 NEXT A:CLS
40 AS=INPUT$(1) 610 VPOKE 2048,0
50 IF AS=CHR$(91) THEN GOSUB 620 END
.. 660 REM'L' 630 '
60 IF AS=CHR$(93) THEN GOSUB 640 ' insert a space in the li
.. 680 REM'J' .. ne
70 PRINT AS; 650 '
80 IF AS <> CHR$(27) THEN 240 660 X=PEEK(&HFA04):Y=PEEK(&HFA
.. 03):P=FNVR(X,Y):P=P-41
90 ' 670 FOR T=(Y+40)+39 TO P+1 STE
00 REM save screen image in A 680 X=PEEK(&HFA04):Y=PEEK(&HFA
.. $ array .. 03):P=FNVR(X,Y):P=P-41
10 ' 690 FOR T=(Y-1)*40 TO P:VPOKE
20 FOR T=932 TO 947:READ AS:V .. T,VPEEK(T+1):NEXT:VPOKE P,
.. 0: RETURN 280
30 DATA *,P,L,E,A,S,E,*,*,*,* 700 'end
.. ,W,A,I,T,*
40 'FOR T=0 TO 800 STEP 40
50 'FOR S=0 TO 39

```